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/* CMP-135 SDTM MH Creation
Domain: MH*/;

options validvarname=upcase nofmterr dlcreatedir;

libname raw "/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1";
libname sdtm "/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm";

libname valxpt xport
"/home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/mh.xpt";

%macro make_date(invar=, outdt=);
  if vtype(&invar) = "N" then do;
    if not missing(&invar) then do;
      if &invar > 100000 then &outdt = datepart(&invar);
      else &outdt = &invar;
    end;
  end;
else do;
  _dtm = input(strip(&invar), anydtdtm.);
  &outdt = datepart(_dtm);

  if missing(&outdt) then
    &outdt = input(strip(&invar), anydtdte.);
end;
%mend;

data ex_prep;
  set raw.ex;

  %make_date(invar=EXSTDN, outdt=exdt);

  format exdt date9.;

  keep SUBJECT exdt;
run;

proc sql;
  create table ex_first as
  select
    SUBJECT,
    min(exdt) as rfstdt format=date9.
  from ex_prep
  where not missing(exdt)
  group by SUBJECT;
quit;

data dov_prep;
  set raw.dov;

  %make_date(invar=VISDTN, outdt=dovdt);

  length INSTANCENAME_U $100 FOLDERNAME_U $100;

  INSTANCENAME_U = upcase(strip(INSTANCENAME));
  FOLDERNAME_U = upcase(strip(FOLDERNAME));

  format dovdt date9.;

  keep SUBJECT INSTANCENAME_U FOLDERNAME_U VISDTN dovdt;
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run;
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proc sql;
  create table mh_with_dov as
  select
    a.*,
    b.dovdt as mh_collection_dt format=date9.
  from raw.mh as a
  left join dov_prep as b
    on a.SUBJECT = b.SUBJECT
    and upcase(strip(a.INSTANCENAME)) = b.INSTANCENAME_U;
quit;
```

```
proc sql;
  create table dov_screen as
  select
    SUBJECT,
    min(dovdt) as screening_dt format=date9.
  from dov_prep
  where FOLDERNAME_U = "SCREENING"
    or INSTANCENAME_U = "SCREENING"
    or index(FOLDERNAME_U, "SCREEN") > 0
    or index(INSTANCENAME_U, "SCREEN") > 0
  group by SUBJECT;
quit;
```

```
proc sql;
  create table mh_merge as
  select
    a.*,
    b.screening_dt,
    c.rfstdt
  from mh_with_dov as a
  left join dov_screen as b
    on a.SUBJECT = b.SUBJECT
  left join ex_first as c
    on a.SUBJECT = c.SUBJECT;
quit;
```

```
data mh_pre;
  set mh_merge;
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```
length
  STUDYID  $6
  DOMAIN  $2
  MHTERM   $64
  MHLLT    $35
  MHDECOD  $40
  MHHLT    $65
  MHHLGT   $86
  MHCAT    $23
  MHBODSYS $67
  MHSOC    $67
  MHDTC    $10
  MHSTDTC  $10
  MHENDTC  $10
  MHENRPT  $7
  MHENTPT  $10
  USUBJID  $40;
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```

STUDYID = "CMP135";
DOMAIN  = "MH";
USUBJID = catx("-", STUDYID, strip(SUBJECT));

MHTERM  = strip(MHTERM);
MHLLT   = strip(MHTERM_LLTT);
MHDECOD = strip(MHTERM_PT);
MHHLT   = strip(MHTERM_HLT);
MHHLGT  = strip(MHTERM_HLGT);

MHLLTCD_N = input(strip(vvalue(MHTERM_LLTT_CODE)), best.);
MHPTCD_N  = input(strip(vvalue(MHTERM_PT_CODE)), best.);
MHHLTCD_N = input(strip(vvalue(MHTERM_HLT_CODE)), best.);
MHHLGTCD_N = input(strip(vvalue(MHTERM_HLGT_CODE)), best.);
MHBDSYCD_N = input(strip(vvalue(MHTERM_SOC_CODE)), best.);
MHSOCCD_N  = input(strip(vvalue(MHTERM_SOC_CODE)), best.);

MHCAT     = "GENERAL MEDICAL HISTORY";
MHBODSYS  = strip(MHTERM_SOC);
MHSOC     = strip(MHTERM_SOC);

if not missing(mh_collection_dt) then mhdtdt = mh_collection_dt;
else mhdtdt = screening_dt;

MHDTC = put(mhdtdt, yymmdd10.);

if not missing(MHSTDN_YY) then do;
    if not missing(MHSTDN_MM) and not missing(MHSTDN_DD) then
        MHDTC = cats(put(MHSTDN_YY, z4.), "-", put(MHSTDN_MM, z2.), "-", put(MHSTDN_DD, z2.));
    else if not missing(MHSTDN_MM) then
        MHDTC = cats(put(MHSTDN_YY, z4.), "-", put(MHSTDN_MM, z2.));
    else
        MHDTC = cats(put(MHSTDN_YY, z4.));
end;

if not missing(MHENDN_YY) then do;
    if not missing(MHENDN_MM) and not missing(MHENDN_DD) then
        MHDTC = cats(put(MHENDN_YY, z4.), "-", put(MHENDN_MM, z2.), "-", put(MHENDN_DD, z2.));
    else if not missing(MHENDN_MM) then
        MHDTC = cats(put(MHENDN_YY, z4.), "-", put(MHENDN_MM, z2.));
    else
        MHDTC = cats(put(MHENDN_YY, z4.));
end;

if not missing(mhdtdt) and not missing(rfstddt) then do;
    if mhdtdt >= rfstdtdt then MHDY = mhdtdt - rfstdtdt + 1;
    else MHDY = mhdtdt - rfstdtdt;
end;

if MHONG = 1 then do;
    MHENRTPT = "ONGOING";
    MHENTPT  = put(screening_dt, yymmdd10.);
end;
else do;
    MHENRTPT = "";
    MHENTPT  = "";
end;

keep
    STUDYID DOMAIN MHTERM MHLLT MHLLTCD_N MHDECOD MHPTCD_N
    MHHLT MHHLTCD_N MHHLGT MHHLGTCD_N MHCAT MHBODSYS MHBDSYCD_N
    MHSOC MHSOCCD_N MHDTC MHSTDTC MHENDTC MHDY MHENRTPT MHENTPT
    USUBJID;

run;

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```

proc sort data=mh_pre out=mh_sorted;
  by STUDYID USUBJID MHTERM MHSTDTC MHENDTC;
run;

data sdtm.mh;
  retain
    STUDYID DOMAIN MHSEQ MHTERM MLLT MLLTCD MHDECOD MHPTCD
    MHLT MHLTCD MHLGT MHLGTC MHCAT MHBODSYS MHBDSYCD
    MHSOC MHSOCCD MHTC MHSTDTC MHENDTC MHDY MHENRTPT MHENTPT
    USUBJID;

  length
    STUDYID $6
    DOMAIN $2
    MHTERM $64
    MLLT $35
    MHDECOD $40
    MHLT $65
    MHLGT $86
    MHCAT $23
    MHBODSYS $67
    MHSOC $67
    MHTC $10
    MHSTDTC $10
    MHENDTC $10
    MHENRTPT $7
    MHENTPT $10
    USUBJID $40;

  set mh_sorted;
  by STUDYID USUBJID;

  if first.USUBJID then MHSEQ = 1;
  else MHSEQ + 1;

  MLLTCD = MLLTCD_N;
  MHPTCD = MHPTCD_N;
  MHLTCD = MHLTCD_N;
  MHLGTC = MHLGTC_N;
  MHBDSYCD = MHBDSYCD_N;
  MHSOCCD = MHSOCCD_N;

  label
    STUDYID = "Study Identifier"
    DOMAIN = "Domain Abbreviation"
    MHSEQ = "Sequence Number"
    MHTERM = "Reported Term for the Medical History"
    MLLT = "Lowest Level Term"
    MLLTCD = "Lowest Level Term Code"
    MHDECOD = "Dictionary-Derived Term"
    MHPTCD = "Preferred Term Code"
    MHLT = "High Level Term"
    MHLTCD = "High Level Term Code"
    MHLGT = "High Level Group Term"
    MHLGTC = "High Level Group Term Code"
    MHCAT = "Category for Medical History"
    MHBODSYS = "Body System or Organ Class"
    MHBDSYCD = "Body System or Organ Class Code"
    MHSOC = "Primary System Organ Class"
    MHSOCCD = "Primary System Organ Class Code"
    MHTC = "Date/Time of History Collection"
    MHSTDTC = "Start Date/Time of Medical History Event"
    MHENDTC = "End Date/Time of Medical History Event"
    MHDY = "Study Day of History Collection"

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MHENRTPT = "End Relative to Reference Time Point"
MHENTPT  = "End Reference Time Point";
/*      USUBJID  = "Unique Subject Identifier"; */

keep
STUDYID DOMAIN MHSEQ MHTERM MHLT MHLTCD MHDECOD MHPTCD
MHLT MHLTCD MHLGT MHLGTC MHCAT MHBODSYS MHBDSYCD
MHSOC MHSOCCD MHTC MHSTDC MHENDTC MHDY MHENRTPT MHENTPT
USUBJID;

run;

proc sort data=sdm.mh out=mh_created;
  by STUDYID USUBJID MHSEQ;
run;

proc sort data=valxpt.mh out=mh_validate;
  by STUDYID USUBJID MHSEQ;
run;

ods rtf file="/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/MH_PROC_COMPARE.rtf";

title "PROC COMPARE: SDTM MH Created vs Validation MH";

proc compare base=mh_validate compare=mh_created listall criterion=0.000001;
  id STUDYID USUBJID MHSEQ;
run;

ods rtf close;
```